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eSurvey

Survey Today, Secure Tomorrow

Google AI for trusted crop-insurance claims

Team Agribid

Team leader: Aniket Dhumal

PS1 • BFSI: Intelligent Risk, Fraud & Financial Experiences

Disasters create a surge of claims and an evidence gap

Claims surge. Evidence is harder to verify.

Thousands of dispersed parcels need assessment at once. Repeated field visits raise costs while genuine farmers wait alongside suspicious claims.



eSurvey combines Gemini-guided surveys, Vertex AI analysis and Earth Engine context in one claim dossier.

Highest spike of claims in insurance industry due to disaster hits lakhs of acres in 1 day, Totally irregulated agriculture market no audit trails , collecting evidences and assessing each claim is humanly impossible so where the Gemini and Vertex AI shines with evidence collection and loss assessment with Google's infra stack

The scale of the verification challenge

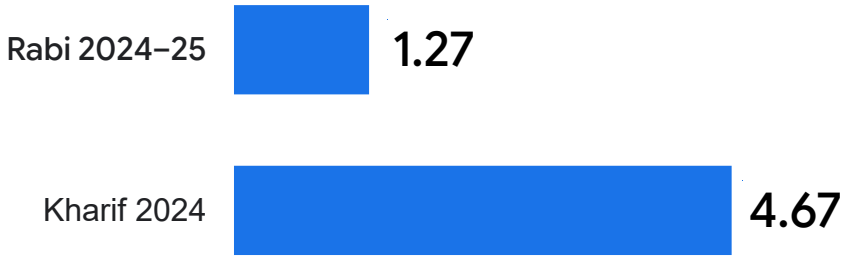
5.94 lakh

Crop-insurance applications rejected in Maharashtra across two seasons

US$308.6bn

Estimated annual US insurance fraud cost
All lines • 2022 study • not India crop losses

Rejected applications (lakh)



Field checks matter

Rejection reasons included barren land and incorrect land / area details.

Sources: Rajya Sabha Q1474, 1 Aug 2025 (4,67,441 + 1,26,830); Coalition Against Insurance Fraud, 2022 study.

One claim, coordinated across Google Cloud

Claim stage	Google technology	Evidence produced
Guide the farmer	Gemini Live API on Vertex AI	Spoken survey guidance and capture prompts
Secure the evidence	Cloud Storage + Pub/Sub	Private media with claim-linked processing events
Coordinate the investigation	Google ADK on Cloud Run	Parallel evidence checks and structured findings
Analyze visible damage	Gemini + custom model on Vertex AI	Crop observations and validated severity estimates
Cross-check the timeline	Google Earth Engine	Parcel-level satellite observations and NDVI history
Prioritize officer review	Cloud SQL + BigQuery	Evidence dossier, decisions and portfolio trends

Gemini Live guides the farmer through the survey



A guided self-survey

Farmer voice + field video

Voice guidance at the parcel

“Show the affected plants, then slowly turn toward the next survey station.”

The app checks location and captures claim-linked clips. Gemini Live provides spoken instructions and requests clearer views when needed.

Local recording and later upload support interrupted connectivity

Underserved market: individual crop-loss verification

₹2.06 lakh crore+

PMFBY claims paid since inception
Through Rabi 2025–26

₹9,837.61 crore

Kharif 2025 claims already paid
Reported by PIB on 29 August 2026

Initial buyer and user

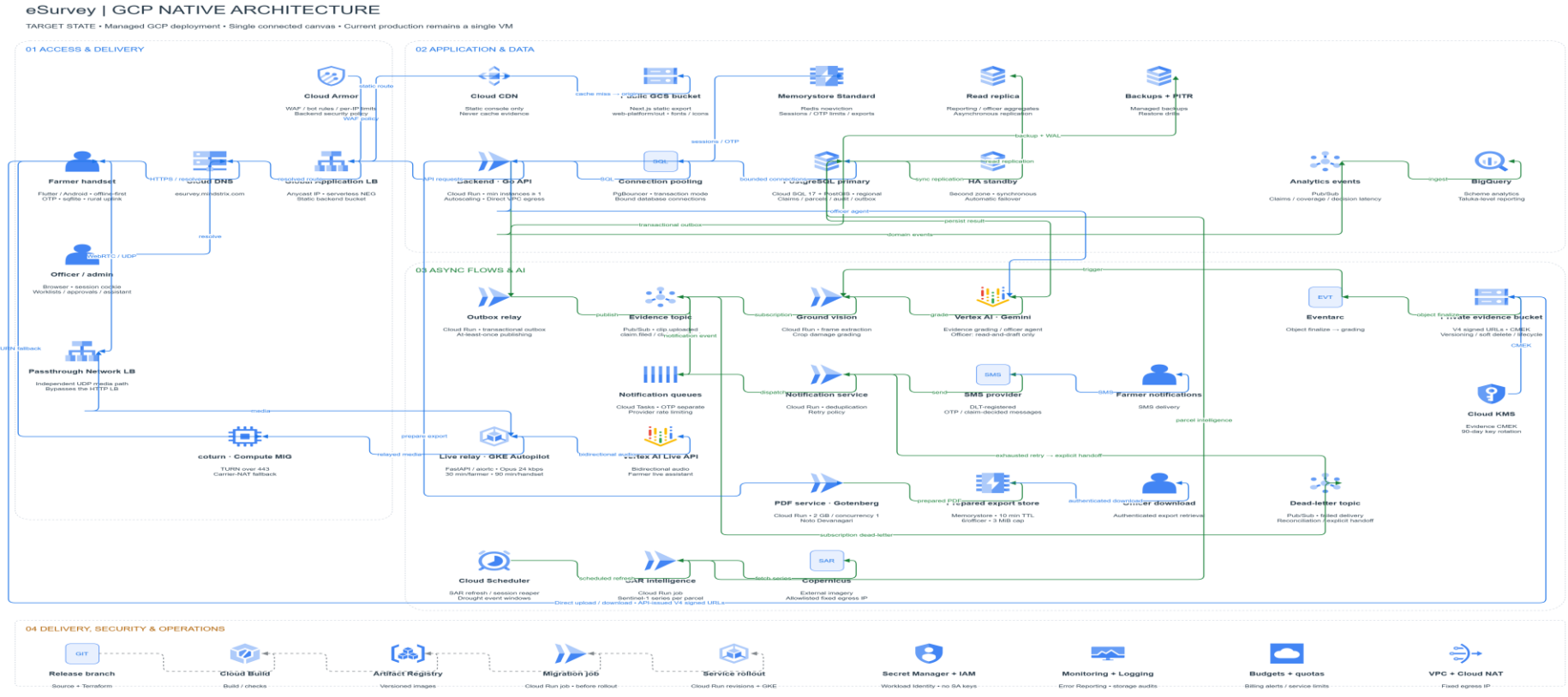
Sell to insurers / loss assessors. Serve farmers who face travel, connectivity and language barriers to documenting a loss.

Our proposed differentiation

Gemini-guided evidence capture plus satellite cross-checks. Give insurers a reviewable dossier while helping farmers document individual losses.

Source: PIB, 29 Aug 2026. Claims paid are market context, not fraud losses or software revenue.

Original GCP architecture: the platform foundation



Original platform design. Gemini Live, ADK and Earth Engine extensions are detailed on slides 4, 5, 8 and 12.

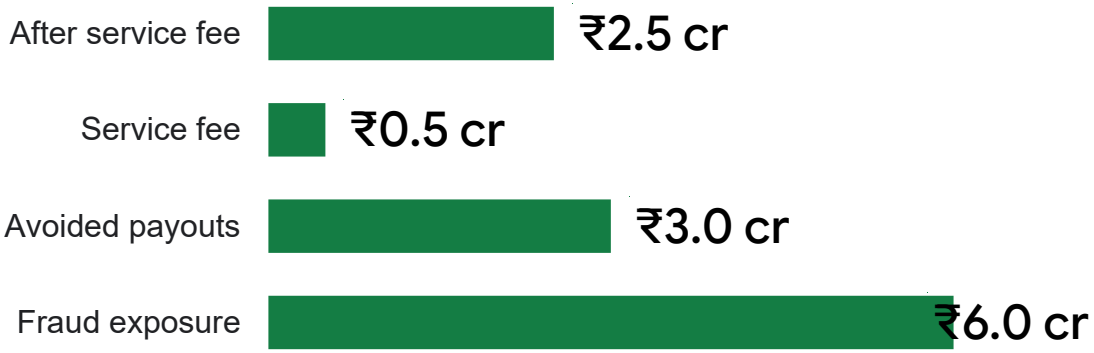
ADK coordinates an evidence-based investigation

Agent / stage	Tool or service	Output for the claim dossier
Intake and quality	Cloud Run + Cloud Storage	Required clips, claim fields and evidence coverage
Visual evidence	Gemini on Vertex AI	Observed crop, visible damage and clip references
Satellite evidence	Earth Engine tool	Dated imagery, NDVI change and quality flags
Record consistency	Cloud SQL / PostGIS tool	Parcel match, duplicate records and declared dates
Evidence synthesis	Gemini + structured output	Agreement, discrepancies and missing information
Decision and learning	Officer review + BigQuery	Recorded disposition and evaluation data

The opportunity: ₹3 crore in potential avoided payouts

₹200 crore claims portfolio

1 lakh claims × ₹20,000 average claim • all inputs below are assumptions



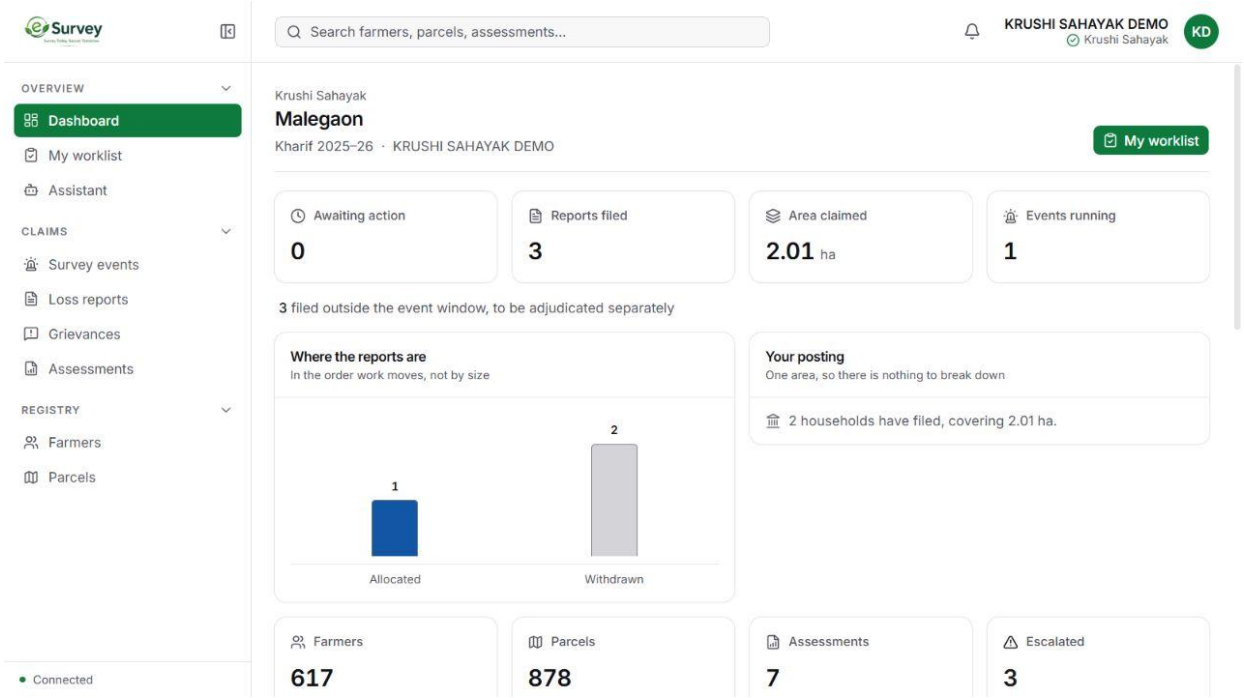
₹3 crore

Potential gross avoided payouts

3% fraud value × 50% prevented
₹50 per claim assumed fee

Illustrative, not achieved savings or a price quote. ₹2.5 crore after the fee excludes investigation, integration and other costs.

eSurvey gives officers one place to investigate claims



Working prototype

Claims worklists
Parcel registry
Assessment routing

Built around the claim officer's review workflow.

Working demo, 18 Oct 2026. End-to-end evidence capture and AI validation are next-round milestones.

Pilot success: better detections, faster claim reviews

Pilot goal: useful fraud signals without delaying genuine claims

Measure	Pilot method	What it demonstrates
Fraud-signal precision / recall	Adjudicated held-out cases; rule-only baseline	Incremental value from Gemini and satellite checks
False refusals / missed fraud	Review clean claims and missed cases	Farmer harm and remaining risk
Confirmed avoided payouts	Officer-confirmed amounts; track appeals	Financial value attributable to eSurvey
Review time / AI cost per claim	Compare existing and assisted workflows	Operational benefit and cost viability

Proposed evaluation on independently adjudicated cases. Compare results with the insurer's existing review process.

Vertex AI and Earth Engine cross-check the same claim

Vertex AI: what is visible?

Gemini describes crop and damage evidence. A custom vision model can estimate severity after training on expert-labeled field data.

Earth Engine: what changed?

Query the parcel across dates. Compare cloud-masked Sentinel-2 NDVI with seasonal history and the reported event window.

ADK: where do findings disagree?

Combine clips, satellite dates and parcel records. Explain conflicts and missing coverage, with links to the underlying evidence.

The insurer: what action follows?

Request another view, prioritize inspection or continue normal review. Record the outcome to improve future evaluation.

Proposed integration. NDVI indicates vegetation change, not cause or payable loss. Cloud cover and revisit timing can limit evidence.

Submission links

GitHub public repository

https://github.com/Aniket-Dhumal/esurvey-builder-cup

Demo video (3-4 minutes)

https://drive.google.com/file/d/1w_wAz0uuPxLQ0Q-vxbzRQ7oo9ejN0PXF/view?usp=sharing

Final product

https://esurvey.mindstrix.com

Why Agribid: Google AI applied to a costly insurance gap

Better evidence for every crop-loss claim.

Gemini helps farmers capture it. Vertex AI and Earth Engine help insurers examine it. ADK brings the findings together.

Value for insurers and farmers

Prioritize suspicious claims, reduce repeated evidence requests and help genuine farmers move through review with clearer documentation.

Next-round demonstration

Show one complete survey-to-dossier journey. Evaluate it with adjudicated claims for detection quality, review time and financial value.

Team Agribid | eSurvey | Working officer prototype with a Google AI investigation workflow

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Thank you

